Antioxidant and Anti-inflammatory Activities of Methanol Extracts of *Tremella fuciformis* and Its Major Phenolic Acids

Hua Li, Hee-Seok Lee, Su-Hwan Kim, BoKyung Moon, and Chan Lee

C: Food Chemistry

Abstract: Methanol extract subfractions of the edible white jelly mushroom (*Tremella fuciformis*), were assessed for the following antioxidant properties: $ABTS^+$ radical scavenging activity, 1,1-diphenyl-2-picrylhydrazyl (DPPH) radical scavenging activity, and inhibitory activity of human low-density lipoprotein (LDL) oxidation. Among the subfractions tested, the chloroform subfraction exhibited the strongest antioxidant activity, with the highest total phenolic content (66.31 μg CAE/mg extract) and flavonoids content (5.12 μg QE/mg extract). The $ABTS^+$ radical scavenging activity of the chloroform subfraction was 7.89 μmol trolox/mg extract, which was the highest among all subfractions. This subfraction also showed the highest DPPH radical scavenging activity and inhibitory activity of LDL oxidation. In addition, the chloroform subfraction demonstrated anti-inflammatory activity through inhibition of nitric oxide production and inducible nitric oxide synthase expression in RAW 264.7 cells. Major phenolic acids from the mushroom extract were identified as 4-hydroxybenzoic acid (323 mg/kg dry weight of mushroom), gentisic acid (174 mg/kg dry weight of mushroom), and 4-coumaric acid (30 mg/kg dry weight of mushroom).

Keywords: anti-inflammatory activity, antioxidant activity, inducible nitric oxide synthase (iNOS), phenolic acid, white jelly mushroom (*Tremella fuciformis*)

Practical Application: Knowledge about the antioxidant and anti-inflammatory activities of *Tremella fuciformis* will promote its usage as a functional food and this mushroom can be utilized as an antioxidant sources in food industry.

Introduction

Oxidative stress induced by free radicals in response to radiation, chemical reactions, and redox reactions has been associated with disorders such as cancer, atherosclerosis, diabetes, and cirrhosis, and may contribute to protein oxidation, DNA damage, and lipid peroxidation in living cells and tissues (Halliwell and Gutteridge 2003). Also, oxidative stress has been related to inflammation, as lipopolysaccharide (LPS)-induced inflammatory factors may be mediated through reactive oxygen species (Park and others 2010). Oxidative/nitrosative stress induces cyclooxygenase-2 (COX-2) and inducible nitric oxide synthases (iNOS, NOS-2), both of which are also found in the inflammatory response caused by bacterial LPS in macrophages (Tsoyi and others 2008). Antioxidants in foods, including vitamin C, tocopherol, carotenoids, phenolic compounds, and flavonoids, might prevent free radical-induced damage (Niki and others 1994). They also may act by inhibiting pro-inflammatory activators to halt the initiation and propagation of the inflammatory response (Park and others 2010).

Mushrooms contain various phenolic compounds widely recognized as excellent antioxidants (Ishikawa and others 1984). The antioxidant activities of several edible mushrooms have been strongly correlated with their total phenolic contents (Cheung and others 2003; Lo and Cheung 2005). Phenolic compounds are one of the major groups of nonessential dietary components that have been associated with the inhibition of atherosclerosis and cancer (Teissedre and others 1996; Williams and Iatropoulos 1997). The bioactivity of phenolics may be related to their ability to chelate metals, inhibit lipoxygenase, and scavenge-free radicals (Decker 1997). Two novel prenylated phenolics, asiaticusin A and B, have been isolated from the fruiting bodies of an edible mushroom, *Boletinus asiaticus* (Wada and others 1996). Another mushroom, *Paxillus panuoides*, was found to contain 2 p-terphenyls that potently inhibit lipid peroxidation (Yun and others 2000). However, there has been limited investigation into antioxidants derived from edible mushrooms that are cultivated artificially.

The soluble extract of *Tremella fuciformis*, one of the most favorite edible mushrooms cultivated artificially in China exhibited various biological activities, including immunomodulatory, antineoplastic, anti-inflammatory functions (Ukai and others 1972; Seljelid 1986; Lin and Cui 2000; Kim and others 2006), and hypoglycemic activities (Kiho and others 1995). The polysaccharide fraction of this mushroom has been also used clinically in China to enhance the immune function of cancer patients receiving chemotherapy or radiotherapy (Gao and others 1997; Lin and Cui 2000). The hot water soluble extract of *T. fuciformis* has been also reported to scavenge superoxide anion (Tsai and others 2003), but the active components remain unknown.

In this study, we investigated the antioxidant and anti-inflammatory effects of edible mushroom, *T. fuciformis*, and identified its active compounds. For this purpose, we measured the antioxidant activities of various subfractions of the methanol extract of *T. fuciformis* and the major phenolic components of the chloroform subfraction, which showed the highest antioxidant

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Table 1–Extraction yields, total phenolic and flavonoid content, and ABTS$^+$ radical scavenging activities of various subfractions of methanol extract of *T. fuciformis*.

Subfraction	Yield (%)[a]	Total phenolic (μg CAE[b]/mg extract)	Total flavonoid (μg QE[c]/mg extract)	TEAC[d] (μmol trolox/mg extract)	FRAP[e] (mmol $FeSO_4 \cdot 7H_2O$/g extract)
n-Hexane	0.46 ± 0.07^{B}	39.07 ± 2.90^{B}	3.86 ± 0.95^{AB}	5.07 ± 0.49^{BC}	ND[f]
Chloroform	0.06 ± 0.01^{B}	66.31 ± 8.46^{A}	5.12 ± 0.88^{A}	7.89 ± 0.08^{A}	61.58 ± 1.23^{A}
Ethyl acetate	0.53 ± 0.07^{B}	25.61 ± 4.70^{B}	4.06 ± 0.32^{AB}	6.54 ± 0.36^{AB}	20.64 ± 0.71^{B}
n-Butanol	3.17 ± 0.21^{A}	8.97 ± 0.82^{C}	2.43 ± 0.16^{BC}	3.47 ± 0.85^{CD}	ND[f]
Water	2.69 ± 0.35^{A}	2.97 ± 0.93^{C}	1.43 ± 0.57^{C}	1.10 ± 1.41^{D}	ND[f]

Data are expressed as means ($n = 3$). Values are average ± standard deviations of 5 different subfractions analyzed in triplicate. Different superscript letters within a column denote statistically significant differences (*$P < 0.05$).
[a]Yield (%) was based on total dry weight of mushroom powder.
[b]Chlorogenic acid equivalents.
[c]Quercetin equivalents.
[d]Trolox equivalent antioxidant capacity.
[e]Ferric reducing/antioxidant power.
[f]Not detected.

activities, were identified using electrospray ionization tandem mass spectrometry (ESI LC-MS/MS). Furthermore, RAW 264.7 cells were then applied to measure the anti-inflammatory activity of the chloroform subfraction *T. fuciformis*.

Materials and Methods

Materials

White jelly mushroom (*T. fuciformis*) was purchased from an open market in Seoul, Korea. All phenolic standards, LPS (L3129), low-density lipoprotein (LDL, L8292) were purchased from Sigma-Aldrich Co., Ltd. (St. Louis, Mo., U.S.A.). RAW 264.7 mouse macrophages were obtained from the Korean Cell Line Bank (Seoul, South Korea). iNOS and β-actin were products of Abcam (Cambridge, UK), and COX-2 was from Santa Cruz Biotechnology (Calif., U.S.A.). Enhanced chemiluminescence reagent was product of GE Healthcare (United Kingdom). High-performance liquid chromatography (HPLC) solvents were obtained from Duksan (Ansan, Gyeonggi-Do, South Korea).

Preparation of samples

Preparation of samples for determination of antioxidant activity. Subfractions of *T. fuciformis* were prepared as follows: *T. fuciformis* mushroom was milled into powder (20 mesh) using a food mixer (HANIL Electric, South Korea) after freeze-dring (EYELA, Rikakikai Co., Ltd, Tokyo, Japan), and 100 g of mushroom powder was extracted twice with 80% methanol (1 L) overnight at room temperature. The extract was filtered through Whatman Nr. 4 filter paper (Whatman International, Kent, UK), and the filtrate was then evaporated to dryness at 40 °C under reduced pressure in a rotary evaporator (EYELA, Rikakikai Co., Ltd). The dried extract was redissolved in 300 mL of 50% methanol, and 300 mL of *n*-hexane was added to the flask. The mixture was then partitioned into hexane and aqueous layers. After separation of the hexane layer, 300 mL of chloroform was added to the aqueous layer, and the chloroform layer was separated from the partitioned solution. The same liquid–liquid partitioning procedure was repeated to obtain ethyl acetate, *n*-butanol, and final aqueous extracts. All extracts were evaporated to dryness and then weighed (Figure 1).

Preparation of phenolic extracts for ESI LC-MS/MS analysis. Phenolic acids were extracted according to the method described by Krygier and others (1982), with minor modifications. We used the 80% methanol fraction from section "Preparation of samples for determination of antioxidant activity" and concentrated to one-fourth of original volume by evaporation under vacuum conditions at 40 °C (EYELA, Rikakikai Co., Ltd). The pH was adjusted to 2 with 6 M HCl and the concentrate was then centrifuged to separate the cloudy precipitate, which was subsequently used for identification of specific phenolic acids.

Evaluation of antioxidant activity

Determination of total phenolic and total flavonoid contents. To measure the total phenolic content of the sample extracts (Singleton and others 1965), a small aliquot (0.5 mL) of each extract sample was mixed with 1.5 mL of distilled water and 0.5 mL of Folin-Ciocalteu reagent. After 5 min of incubation at room temperature, 0.5 mL of Na_2CO_3 (0.1 g/mL

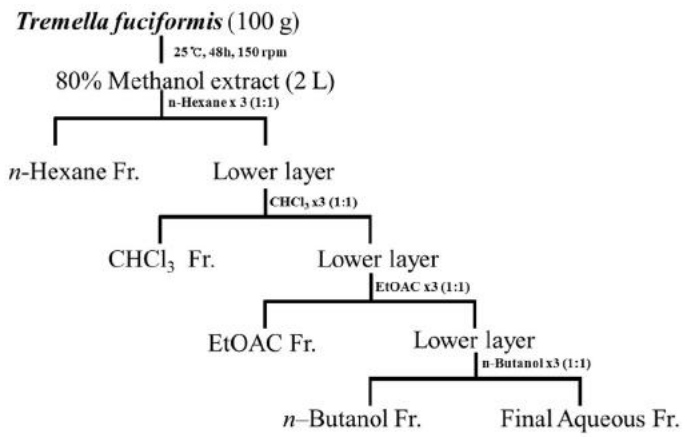

Figure 1–Preparation of *T. fuciformis* subfractions by systemic extraction.

Table 2–Phenolic acids identified by ESI LC-MS/MS comparison with standards.

Compound	$CHCl_3$[a] (μg/g)[d]	EtOAC[b] (μg/g)[d]	Dry weight[e]	R.T.[c]	MS(m/z) (M-H)	MS/MS (m/z)
4-Hydroxybenzoic acid	22.1 ± 0.50	58.2 ± 3.24	323 ± 7.56	5.88	137	93
Gentisic acid	90.7 ± 0.16	22.5 ± 0.26	174 ± 2.54	5.95	153	109
4-Coumaric acid	9.5 ± 0.44	4.5 ± 0.11	30 ± 2.10	8.95	163	119

Data are expressed as means ± standard deviation ($n = 3$).
[a]Chloroform subfraction.
[b]Ethyl acetate subfraction.
[c]Retention time.
[d]μg/g extract.
[e]mg/Kg dry weight of mushroom = total (chloroform + ethyl acetate).

aqueous solution) was added to the mixture. The mixture was kept in the dark for 60 min, and its absorbance was measured at 640 nm using a UV-visible spectrophotometer (Uvikon 933, Milan, Italy). The phenolic content in each sample was expressed as μg chlorogenic acid equivalents per milligram of extract.

To determine the total flavonoid content (Moreno and others 2000), the dried extracts were dissolved in 1 mL of 80% ethanol, and 100 μL of the sample was mixed with 100 μL of aluminum nitrate (0.1 g/mL aqueous solution), 100 μL of 1 μM potassium acetate, and 4.7 mL of 80% ethanol. After 40 min of incubation at room temperature, the absorbance of the solution was measured at 640 nm using a UV-visible spectrophotometer (Uvikon 933). The results were expressed as μg quercetin equivalents per mg of extract.

Determination of ABTS$^+$ radical scavenging activity. The 2, 2′-azino-di-(3-ethylbenzthiazoline sulphonate) (ABTS)$^+$ reagent (7 mM ABTS with 2.45 mM $K_2S_2O_8$) was mixed with 10 μL of sample in dimethyl sulfoxide solution. The 6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid (trolox) was prepared in the concentration range from 0.5 to 2 mM as a standard. The antioxidant activities of the samples were expressed as trolox equivalent antioxidant capacity (TEAC), which represents the micromolar concentration of trolox having the same activity as 1 mg of sample (Re and others, 1999). After the mixture was kept in the dark at room temperature for 12 min to allow for complete

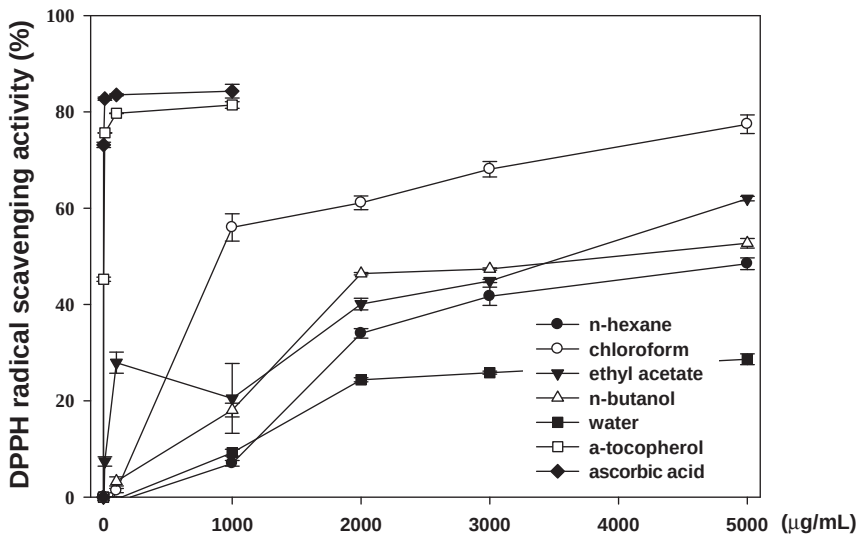

Figure 2–DPPH radical scavenging effects of various subfractions of the methanol extract of *T. fuciformis*.

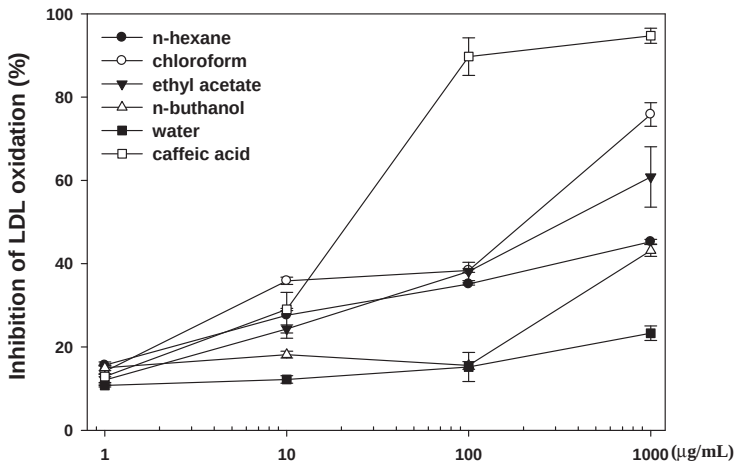

Figure 3–Inhibition of human LDL oxidation by subfractions of the methanol extract of *T. fuciformis*.

free radical generation, the absorbance was measured at 734 nm using a spectrophotometer (Uvikon 933).

Ferric reducing/antioxidant power (FRAP) assay

The FRAP reagent was prepared as follows: 300 mM acetate buffer (pH 3.6), 10 mM 2,4,6-tripyridyl-s-triazine dissolved in 40 mM HCl, and 20 mM $FeCl_3 \cdot 6H_2O$ were mixed in a ratio of 10:1:1 (Alvarez-Parrilla and others, 2007). The mixed FRAP reagent was preheated at 37 °C before use. Note that 24 μL of sample dissolved in methanol and 180 μL of FRAP reagent were mixed in 96 well plate and incubated at 37 °C for 60 min to measure the absorbance at 595 nm. $FeSO_4 \cdot 7H_2O$ was used as the standard and the concentration was 100 to 1000 μM, methanol was the blank. The FRAP was expressed as the concentration of ferric reducing power in 1 g of sample (μmol/g).

Determination of 1,1-diphenyl-2-picrylhydrazyl (DPPH) radical scavenging activity

Sample extracts (100 μL) redissolved in methanol at various concentrations (1, 10, 100, 1000, 2000, 3000, and 5000 μg/mL) were mixed with 100 μM DPPH solution (in 900 μL methanol). The mixture was shaken vigorously and then left to stand for 10 min under subdued lighting. The absorbance was then measured at 518 nm (Chung and others 2005).

Inhibition of human LDL oxidation

The extent of LDL oxidation was determined by measuring the production of thiobarbituric acid reactive substances (TBARS) as described by Zhang and others (1997). A 100 μL aliquot of diluted LDL (5 mg/mL) was incubated with 5 μL of 1 μM $CuSO_4$ and 10 μL of *T. fuciformis* subfractions (0.5 mg/mL methanol) or ultrapure water (negative control) for 24 h at 37 °C. The reaction was stopped by adding 25 μL of ethylenediaminetetraacetic acid (10 mg/mL aqueous solution), after which the mixture was cooled to 4 °C. TBARS was formed by adding 2 mL of TCA–TBA–HCl solution, followed by heating at 95 °C for 1 h. After the mixture had cooled, formation of TBARS was determined by measuring the absorbance at 532 nm. The production of TBARS was expressed in nanomoles of malondialdehyde per milligram of LDL. Caffeic acid (0.5 mg/mL) was used as a positive control.

Identification and quantification of phenolic acids by ESI LC-MS/MS analysis

Twelve phenolic standards and sample (10 μL) were injected into an HPLC system and separated using a reverse-phase C18 column (Union UK-C18, 100 × 2.0 mm, i.d., 3 μm, Imtakt, Japan). The mobile phase consisted of 0.1% aqueous formic acid (solvent A) and acetonitrile solution containing 0.1% formic acid (solvent B). The nonlinear gradient elution used was as follows: 5% B at 0 min, 30% B at 12 min, 30% B at 30 min, 50% B at 32 min, 95% B at 34 min, 95% B at 36 min, 5% B at 38 min, and 5% B at 40 min. The flow rate was 200 μL/min and the capillary temperature was 250 °C. The level of phenolic acids were determined by ESI LC-MS/MS analysis using the same ESI mass spectrometry system (LTQ Velos, Thermo Scientific, San Josh, Calif., U.S.A.) in negative mode (−10 V).

Anti-inflammatory activity of *T. fuciformis* chloroform subfraction

Cell viability. RAW 264.7 cells (1 × 10^5 cell/mL) were incubated in a 96-well plate with fetal bovine serum-free Dulbecco's modified eagle medium containing the *T. fuciformis* chloroform

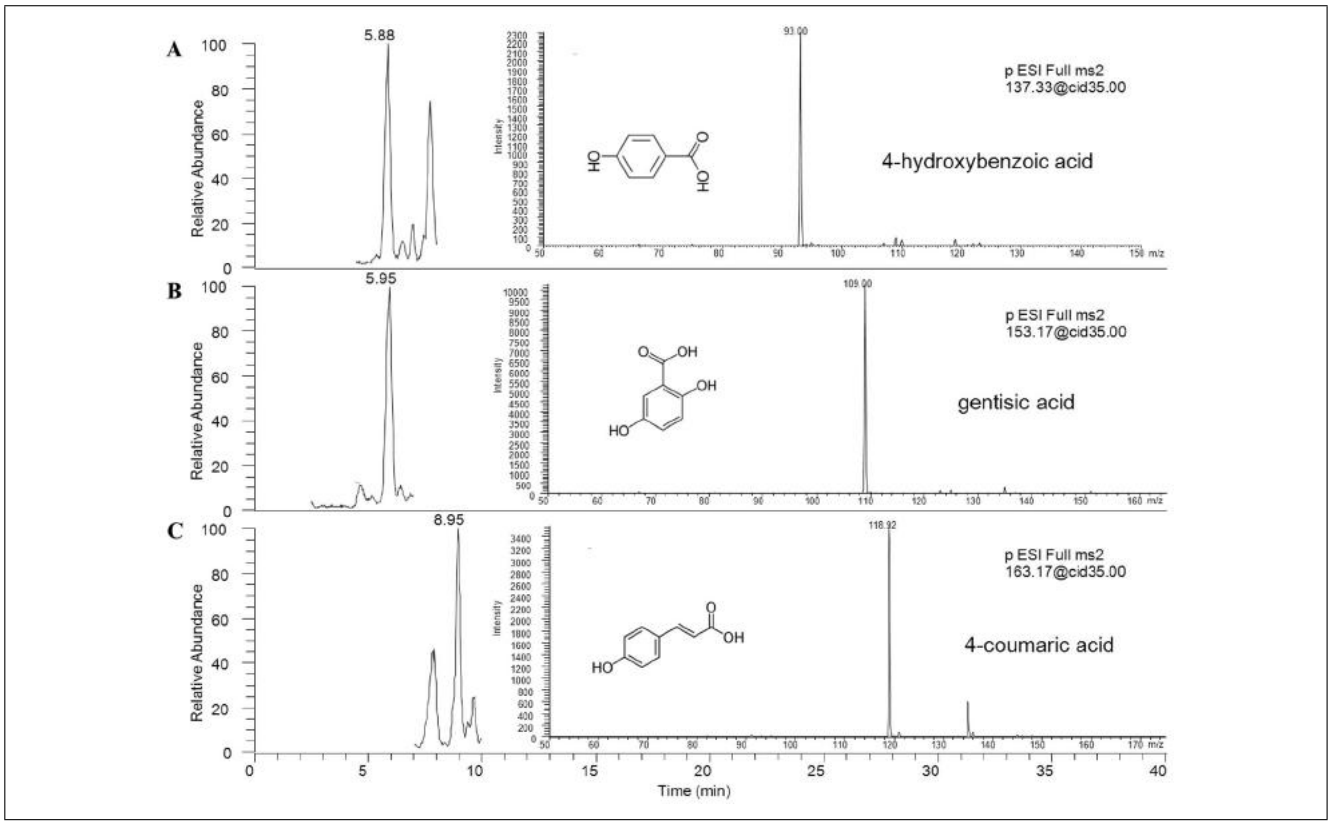

Figure 4–Mass spectrum of the chloroform subfraction. (A) MS/MS analysis of 4-hydroxybenzoic acid in sample. (B) gentisic acid. (C) 4-coumaric acid.

subfraction (25, 50, 100, 200 μg/mL) or identified 3 phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid; 50, 100, 200, 500, and 1000 μg/mL) for 24 h. The 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT, a yellow tetrazole) reagent was used to induce the purple formazan in living cells. The absorbance of this colored solution was quantified by a spectrophotometer (550 nm).

Nitric oxide assay. RAW 264.7 cells were pretreated with chloroform subfraction of *T. fuciformis* at different concentrations (25, 50, 100, and 200 μg/mL) 1 h before LPS administration (1 μg/mL). The effect of identified 3 phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid) on nitric oxide formation was tested at 50, 100, 200, and 500 μg/mL. After addition of 100 μL of Griess reagent to 100 μL of LPS-treated cell culture medium, the absorbance of solution was measured at 550 nm (Tsoyi and others 2008). Sodium nitrite (25~100 μM) was used as standard substance.

Western blotting. The *T. fuciformis* chloroform subfractions (50, 100, and 200 μg/mL) or 3 phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid; 50, 100, 200, and 500 μg/mL) were added to the RAW 264.7 cells 1 h before treatment with LPS (1 μg/mL; 18 h for iNOS expression and 4 h for COX-2 expression). Cells were lysed and total cellular protein was solubilized using radioimmunoprecipitation assay buffer containing protease inhibitor and phosphatase inhibitor. The concentration of protein was measured according to the Bradford protein assay described by Zor and Selinger (1996). An 8% acrylamide gel and polyvinylidene fluoride membrane (0.45 μm, Millipore Corporation, Billerica, Mass., U.S.A.) were used to separate and transfer protein. Skim milk solution (5%, w/w) was used as the blocking agent.

Statistical analysis

All experiments were performed in triplicate, and the values were expressed as mean ± standard deviation. All of the data were subjected to one-way analysis of variance (Scheffe test) to compare the antioxidant activity, phenolic content and flavonoid content of each sample. Values were considered significant at $P < 0.05$.

Results and Discussion

Extraction yields and antioxidant contents

As shown in Table 1, most components in methanol extract were assigned to the *n*-butanol and water subfractions. The *n*-hexane, chloroform, and ethyl acetate subfractions exhibited high content of total phenolic compounds. The *n*-butanol and water subfractions contained comparatively smaller amounts of phenolic compounds. Flavonoids were found in all of the subfractions in the following order: chloroform > ethyl acetate > *n*-hexane > *n*-butanol > water. It has been reported that the antioxidant

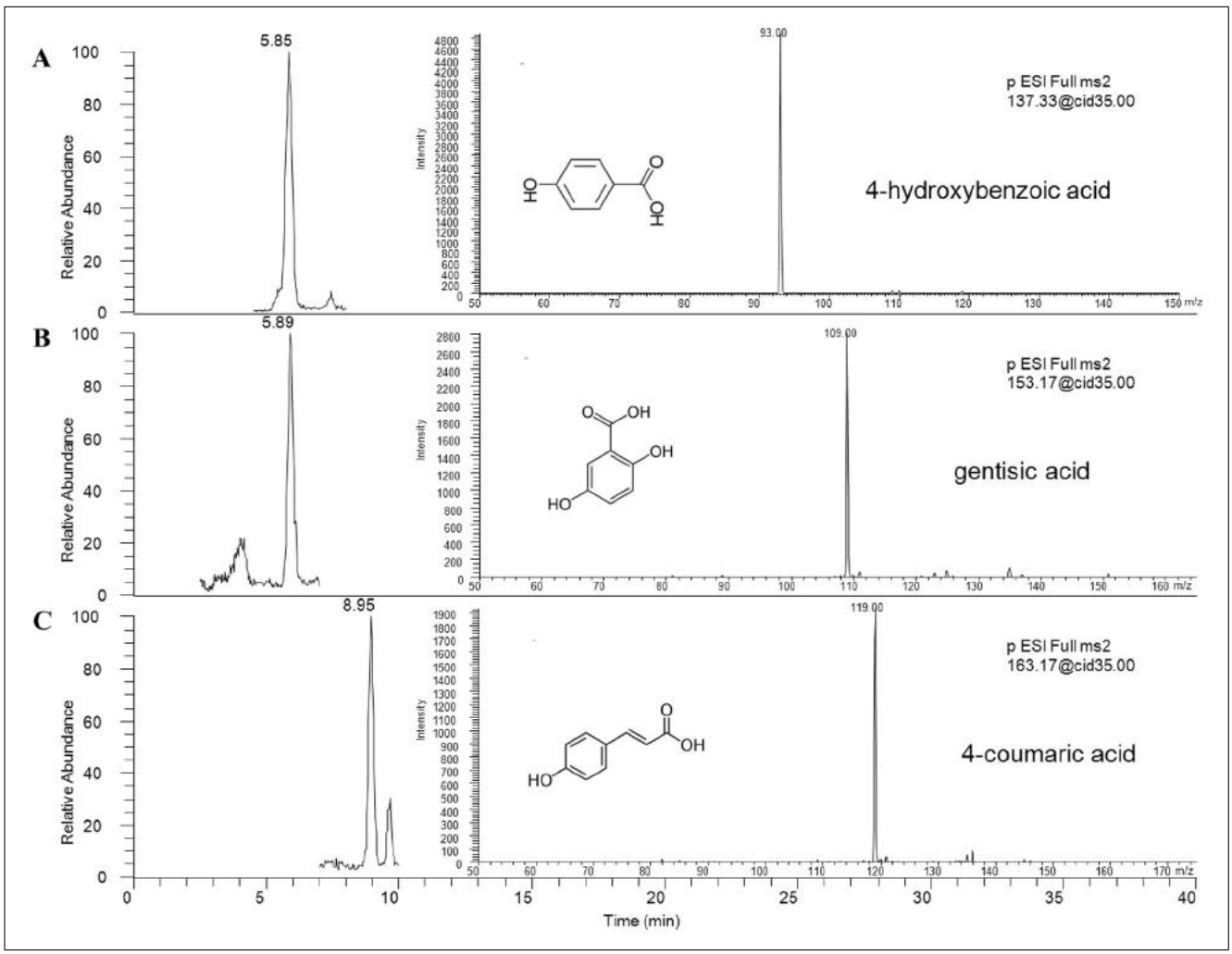

Figure 5–Mass spectrum of the ethyl acetate subfraction. (A) MS/MS analysis of 4-hydroxybenzoic acid in sample. (B) gentisic acid. (C) 4-coumaric acid.

activity of plant material is strongly correlated with the phenolic content (Velioglu and others 1998), and accordingly, the effect of total phenolic content on the antioxidant activities of the mushroom extracts should be considered. The chloroform subfraction of *T. fuciformis*, exhibited larger amount of total phenolic content than methanol extract or its subfractions of *A. aegerita* reported by Lo and Cheung (2005), who explained the correlation between phenolic content and the antioxidant activity in mushroom. Therefore, the intrinsic antioxidant activities of the chloroform subfraction of *T. fuciformis* may be related to phenolic content in extracts.

ABTS$^+$ radical scavenging activity

The antioxidant activities of the methanol extract of *T. fuciformis* were evaluated (Table 1). The TEAC value was the highest for the chloroform subfraction. The hydrophobic subfractions, including chloroform, ethyl acetate, and *n*-hexane, showed higher antioxidative activities than the hydrophilic subfractions such as water (Table 1). This finding suggests that the compounds responsible for the antioxidative properties of the methanol crude extract of *T. fuciformis* were mostly extracted into the hydrophobic subfractions. Among the subfractions, the chloroform subfraction showed the highest level of phenolic compounds, exhibiting correlation with antioxidant potency ($R^2 = 0.8725$; TPC compared with TEAC; $P < 0.05$). Thetsrimuang and others (2011) reported that crude polysaccharides of *L. polychrous* Lev. showed much lower ABTS$^+$ radical scavenging activity (0.0534 to 0.131 μmol trolox/mg extract in mycelia and fruiting bodies) compared with those of methanol extract from *T. fuciformis*.

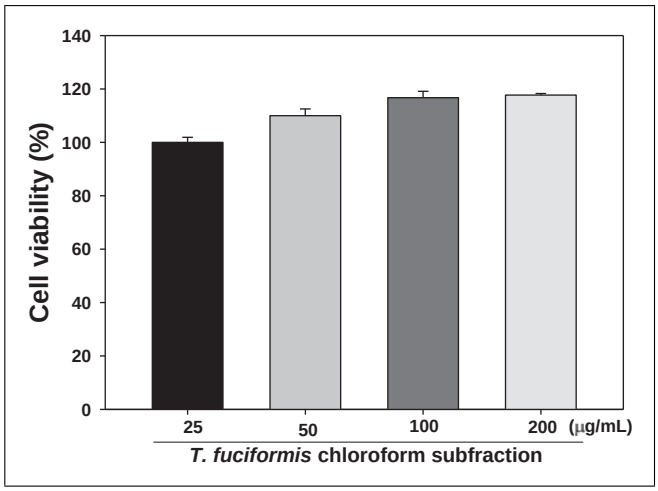

Figure 6–Cell viability of RAW 264.7 mouse macrophage against the *T. fuciformis* chloroform subfraction (TFCS). RAW 264.7 cells were treated with LPS (1 μg/mL) and TFCS (25, 50, 100, 200 μg/mL) for 24 h. Cell viability was determined as described in Materials and Methods.

FRAP assay

FRAP assay is to measure the reduction of Fe^{3+} complex to Fe^{2+} in the presence of antioxidants in samples. The results were shown in Table 1, in which FRAP was expressed as millimoles of $FeSO_4 \cdot 7H_2O$ per gram (extract). FRAP could be analyzed only in the chloroform and ethyl acetate subfractions, and other subfractions did not exhibit FRAP. Wong and others (2009) reported that methanolic extract of fruit body and mycelium of edible mushroom (*Hericium erinaceus*) showed the FRAP value of 4.06 μmol/g and 21.93 μmol/g (dry weight), respectively. The methanol extract of *T. fuciformis* exhibited higher FRAP value (146.34 μmol/g dry weight) than those in the report of Wong and others (2009).

DPPH radical scavenging activity

Among the subfractions tested, the chloroform subfraction exerted the strongest scavenging of DPPH radicals (56% and 77.4% at 1 and 5 mg/mL, respectively; Figure 2). Other subfractions exerted much lower DPPH radical scavenging activities than the chloroform subfraction (less than 20% at 1 mg/mL). The half maximal inhibitory concentration (IC_{50}) is represents the concentration of sample that is required for 50% scavenging. IC_{50} values were 0.8, 3.3, 3.6, 3.8, and 20.3 mg/mL for the chloroform, *n*-hexane, ethyl acetate, *n*-butanol, and water subfractions, respectively. Mau and others (2004) reported that the radical scavenging activities of methanol extracts of *Termitomyces albuminosus* (termite mushroom), *Grifola frondosa* (maitake), and *Morchella esculenta* (morel)

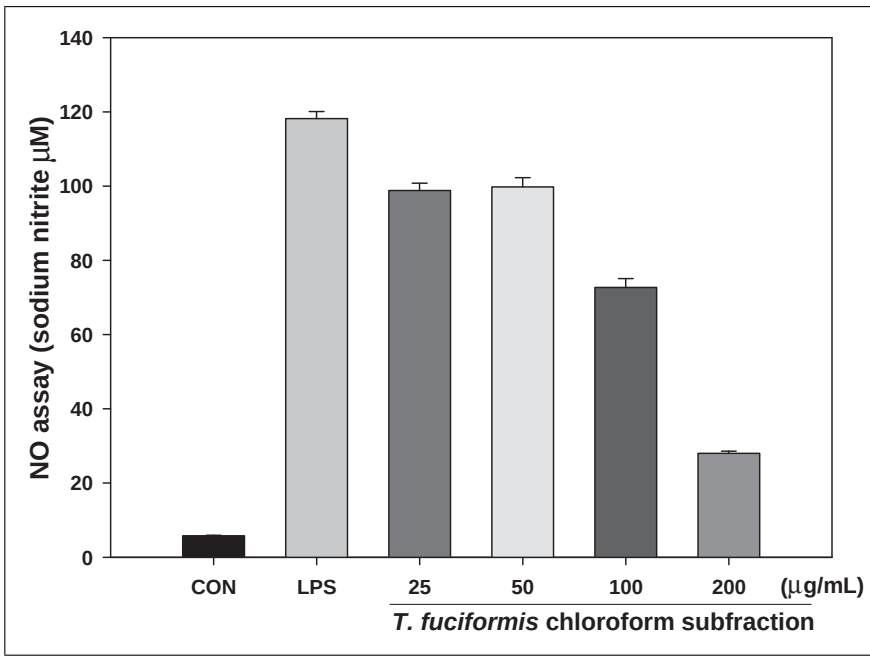

Figure 7–Effect of the *T. fuciformis* chloroform subfraction on nitric oxide production. RAW 264.7 cells were treated with LPS alone (1 μg/mL), or with LPS (1 μg/mL) and TFCS (25, 50, 100, 200 μg/mL) for 24 h. The amount of NO produced was determined as described in Materials and Methods.

were 78.8%, 79.4%, and 94.1%, respectively, at 10 mg/mL. At concentrations below 10 mg/mL, the methanol extract of *M. esculenta* demonstrated the highest DPPH radical scavenging activity among the 3 mushrooms. Huang (2000) found that the methanol extracts of mycelia of *Antrodia camphorata* and *Agaricus blazei* possessed DPPH radical scavenging activities of 97.1% and 98.8%, respectively, at 5 mg/mL. Tsai and others (2003) determined that the methanol extracts of mycelia of *Agrocybe cylindracea* and *Ganoderma tsugae* exerted low radical scavenging effects, measured at 91.4% and 95.6%, respectively, at 10 mg/mL. Therefore, the chloroform subfraction of *T. fuciformis* showed a little bit lower antioxidant activity compared with those of methanol extracts from mycelia of *Antrodia camphorata* and *Agaricus blazei* reported by Huang (2000).

Inhibition of lipid peroxidation using TBARS

LDL oxidation has been implicated in the development of atherosclerosis (Steinberg 1997); thus, the antioxidant effects of various mushroom subfractions on TBARS production were examined by incubating each with human LDL in the presence of 5 μM $CuSO_4$ as an oxidation initiator. The inhibition of LDL oxidation by the extracts is shown in Figure 3. The chloroform subfraction exhibited the strongest antioxidant activity among those tested (38.4% and 75.8% inhibition at 0.1 and 1 mg/mL, respectively). This was followed by the ethyl acetate subfraction at 1 mg/mL (60.2% inhibition), whereas the other extracts showed inhibition effects of less than 50% at the same concentration. The IC_{50} values for inhibition of LDL oxidation by the chloroform, ethyl acetate, and *n*-hexane subfractions were 0.37, 0.75, and 1.2 mg/mL, respectively. Cheung and others (2003) reported that the IC_{50} values for the inhibition of lipid peroxidation in rat brain homogenate by the dichloromethane subfractions of *Lentinus edodes* and *Volvariella volvacea* were 0.297 and 0.109 mg/mL, respectively. Lo and Cheung (2005) found that after 4 h of incubation, the inhibition of LDL oxidation by the ethyl acetate subfraction of *A. aegerita* (0.5 mg/mL) and caffeic acid (0.5 mg/mL) were 3.09 ± 0.38 and 2.69 ± 0.16 mg/mL, respectively. Therefore, lipid peroxidation activity of the chloroform subfraction of the methanol extract of *T. fuciformis* exhibited lower lipid peroxidation activity than those previously measured using extracts of *L. edodes* and *V. volvacea*. The chloroform subfraction showed the highest level of phenolic compounds as well as the strongest antioxidant activity among the tested subfractions.

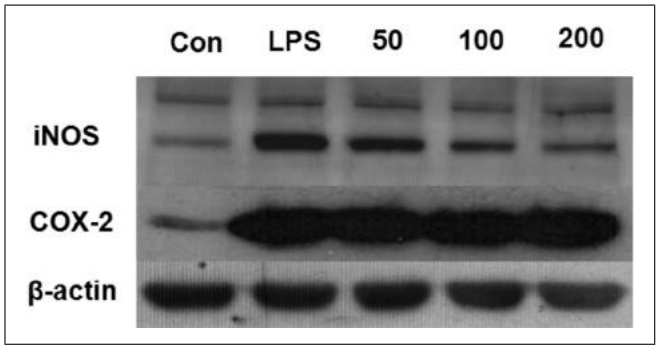

Figure 8–Western blot experiment of expressed iNOS and COX-2 following treatment with LPS and *T. fuciformis* chloroform subfraction. RAW 264.7 cells were treated with LPS alone (1 μg/mL), or with LPS (1 μg/mL) and TFCS (50, 100, 200 μg/mL) for 18 h (iNOS) and 4 h (COX-2).

Identification and quantification of phenolic acids by ESI LC-MS/MS analysis

Phenolic compounds are one such class of metabolites, and have been shown to be excellent antioxidants (Ishikawa and others 1984). In addition, Cheung and others (2003) reported that phenolic compounds made a significant contribution to the antioxidant activities of edible mushroom extracts (*L. edodes* and *V. volvacea*). The major phenolic acids in every subfraction were profiled by comparison of their ESI LC-MS/MS spectra with the standards (Figure 4 and 5). Gentisic acid, 4-hydroxybenzoic acid, and 4-coumaric acid were identified in both chloroform and ethyl acetate subfractions (Table 2). No phenolic acids were detected in the other subfractions (*n*-hexane, *n*-butanol, and water).

Following measurement with mass spectroscopy, total levels of phenolic acids in the fruiting body of *T. fuciformis* were calculated to be 323, 174, and 30 μg/g fruiting bodies for 4-hydroxybenzoic acid, gentisic acid, and 4-coumaric acid, respectively. Barros and others (2009) reported that 4-hydroxybenzoic acid (29.66 μg/g), vanillic acid isomer (4.92 μg/g), and another vanillic acid isomer (7.81 μg/g) were present in the *Tricholoma acerbum* mushroom.

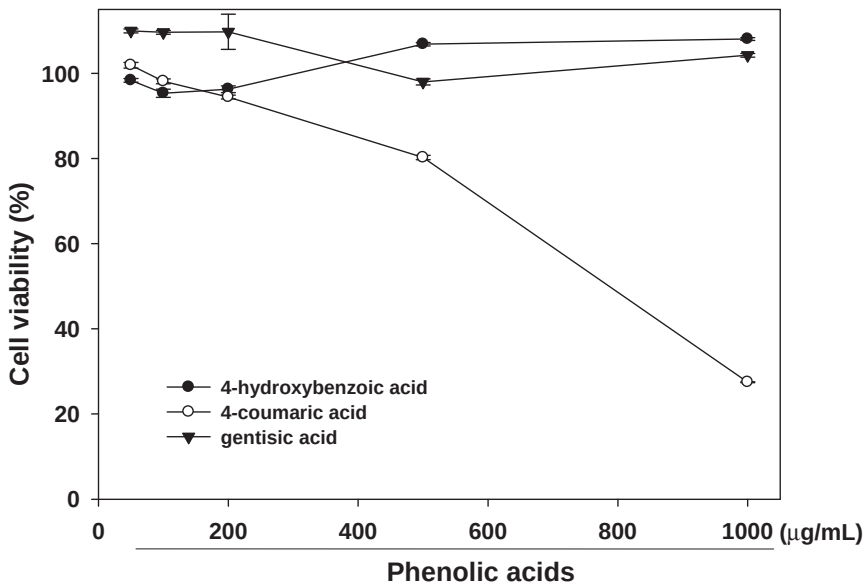

Figure 9–Cell viability of RAW 264.7 mouse macrophage against the 3 phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid). RAW 264.7 cells were treated with LPS (1 μg/mL) and phenolic acids (50, 100, 200, 500, and 1000 μg/mL) for 24 h. Cell viability was determined as described in Materials and Methods.

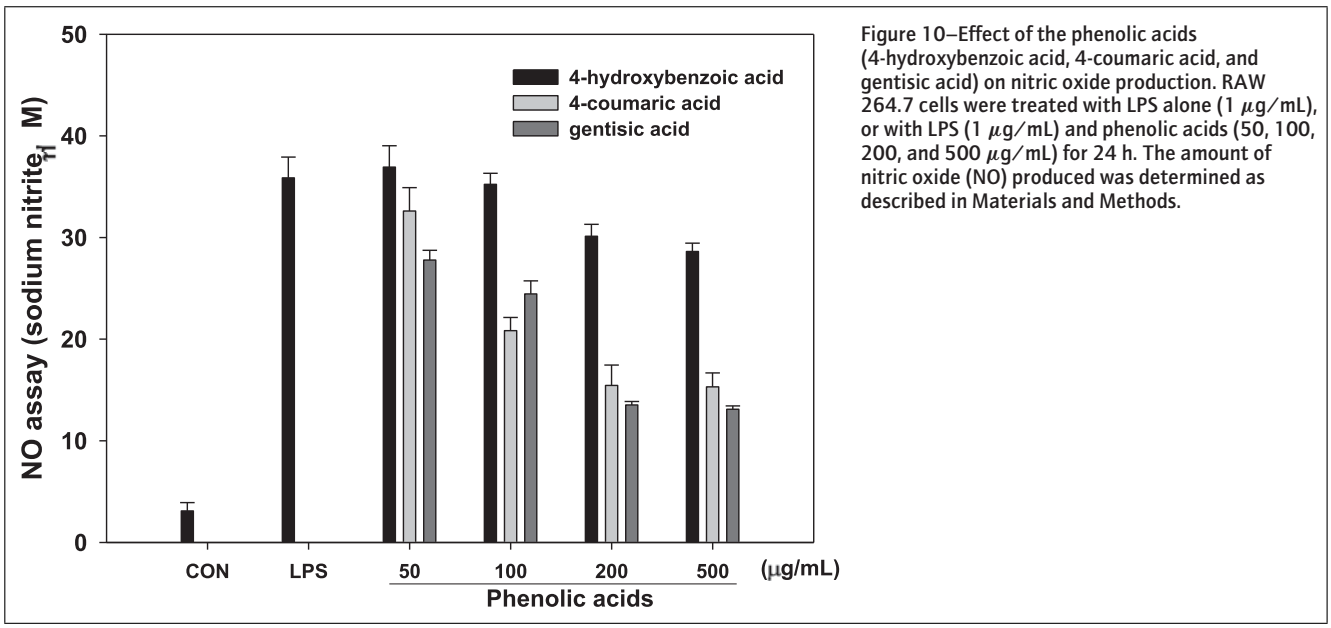

Figure 10–Effect of the phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid) on nitric oxide production. RAW 264.7 cells were treated with LPS alone (1 μg/mL), or with LPS (1 μg/mL) and phenolic acids (50, 100, 200, and 500 μg/mL) for 24 h. The amount of nitric oxide (NO) produced was determined as described in Materials and Methods.

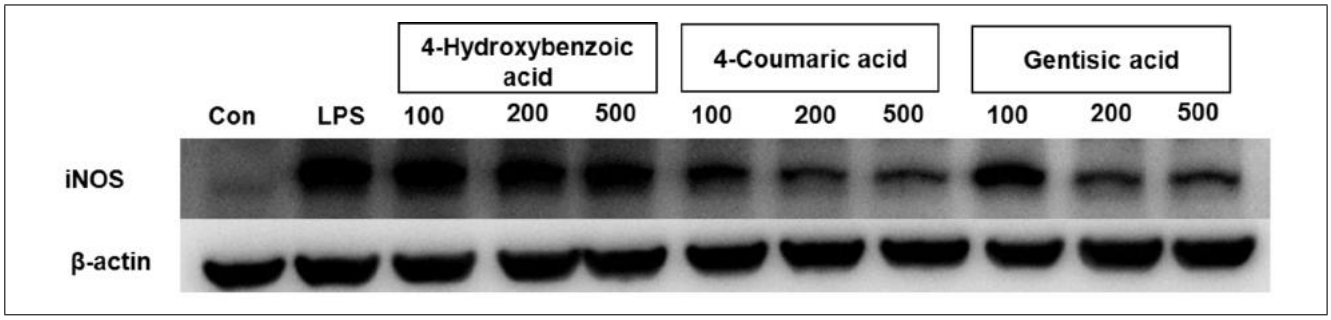

Figure 11–Western blot experiment of expressed iNOS following treatment with LPS and 3 phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, gentisic acid). RAW 264.7 cells were treated with LPS alone (1 μg/mL), or with LPS (1 μg/mL) and phenolic acids (100, 200, 500 μg/mL) for 18 h (iNOS).

In addition, protocatechuic acid (33.47 μg/g), 4-hydroxybenzoic acid (29.31 μg/g), and 4-coumaric acid (3.75 μg/g) were present in the *Lepista nuda* mushroom. Ju and others (2010) reported that some phenolic acids such as vanillic acid, isomer of gentisic acid (protocatechuic acid), and syringic acid were present in the *Inonotus obiquus* mushroom.

Anti-inflammatory activity of *T. fuciformis* chloroform subfraction via nitric oxide production and iNOS and COX-2 expression

As shown in Figure 6, the concentration range chloroform subfraction (25 to 200 μg/mL) exhibiting the highest antioxidant activity did not affect the cell viability of RAW 264.7 mouse macrophages.

The production of nitric oxide decreased in a concentration-dependent manner (Figure 7). The nitric oxide concentration of the control was 5.8 μM, and increased to 118.2 μM with LPS treatment. Interestingly, addition of the *T. fuciformis* chloroform subfraction (200 μg/mL) decreased the level of nitric oxide to 28 μM (80.2% inhibition). Similar effects were reported with other edible mushrooms, such as *C. cibarius*, *C. cornucopioides*, *L. deliciosus*, and *A. bisporus*; treatment with their methanolic extracts at 0.5 mg/mL inhibited nitric oxide production by 30% to 70% (Moro and others 2012).

In general, whole mushroom extract can inhibit LPS-induced iNOS and COX-2 expression in RAW 264.7 mouse macrophages (Park and others 2010). In this experiment, LPS-induced iNOS expression was decreased in a dose-dependent manner by treatment with the chloroform subfraction, similar to the results observed with nitric oxide production. These results suggest that the *T. fuciformis* chloroform subfraction may inhibit nitric oxide production through suppression of iNOS expression in LPS-induced RAW 264.7 cells. Interestingly, COX-2 expression was not reduced with administration of the chloroform subfraction (Figure 8). These results indicate that the chloroform subfraction may have an effect on the signal transducer and activator of transcription (STAT) pathway in order to inhibit iNOS expression as observed. Previous studies have reported similar results (Tsoyi and others 2008) through suppression of STAT signaling.

Three phenolic acids (4-hydroxybenzoic acid, 4-coumaric acid, and gentisic acid), which were identified in chloroform, were tested for their anti-inflammatory activity. The 4-hydroxybenzoic acid and gentisic acid did not exhibit any cytotoxicity at all tested concentrations. However, the 4-coumaric acid showed strong cytotoxicity in which only 27% of the cells were viable at 1000 μg/mL. The 80% of the cells viability could be observed at 500 μg/mL (Figure 9). The production of nitric oxide decreased in a concentration-dependent manner by addition of 4-coumaric acid and gentisic acid (Figure 10). These results indicate that the

4-coumaric acid and gentisic acid strongly inhibit nitric oxide production at 200 μg/mL. As shown in Figure 11, 2 phenolic acids inhibited iNOS expression leading to decrease of nitric oxide production that contributed anti-inflammatory activity in *T. fuciformis*. Similar result was reported by Rasadah and others (2005) using protocatechuic acid, an isomer of gentisic acid for anti-inflammatory activity. Interestingly, 4-hydroxybenzoic acid exhibited very low reduction of iNOS expression and nitric oxide production.

Conclusions

The *T. fuciformis* chloroform subfraction displayed the highest antioxidant activity in all experimental settings and 3 phenolic acids (4-hydroxybenzoic acid, gentisic acid, and 4-coumaric acid) were identified in the chloroform and ethyl acetate subfractions. The chloroform subfraction also exhibited anti-inflammatory activity in LPS-induced RAW 264.7 cells by inhibiting expression of inducible nitric oxide synthase, and thereby suppressing nitric oxide production.

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